Suvinava Basak

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Research Interests

Statistical machine learning researcher focused on the mathematical and empirical foundations of deep learning, with interests in generalization, optimization, uncertainty quantification, calibration and learning under distribution shift. Current work combines theoretical analysis of neural-network behavior with computational experiments, complemented by applications to biomedical machine learning.

Education

Oct '23 – · · · · **M.Sc. Data Science,** *Technische Universität Braunschweig, Germany*
Current average: **1.3** (German scale, 1.0 = best), excluding thesis (due date: Oct 15, 2026).
Thesis: *End-to-End Calibration: Theory and Empirical Evaluation in Biomedical Context*

Jul '16 – Mar '18 **Master of Business Administration,** *Indian Institute of Management Indore, India*

Aug '10 – Jun '14 **B.Tech. Computer Science & Engineering,** *Jalpaiguri Government Engineering College (West Bengal University of Technology), India*
DGPA: **7.83**/10.
Project: *Face Recognition System using MATLAB*

Research Experience

Apr '26 – · · · · **(M.Sc. Thesis) End-to-End Calibration: Theory and Empirical Evaluation in Biomedical Context —**
TU Braunschweig, Institute for Statistics and Foundations of Machine Learning, Germany

• Investigating end-to-end calibration of neural networks, incorporating calibration objectives directly into the model training rather than post-hoc calibration, and empirically evaluating its effect on predictive reliability and uncertainty estimation; supervised by Prof. Dr. Nicole Mücke.

• Studying the relationship between predictive performance, confidence and reliability under distribution shift, using Transformer-based gene regulatory network inference as application.

• Designing and implementing Python/PyTorch experiments to evaluate neural-network calibration methods on biological regulatory network data.

Aug '25 – · · · · **Student Hilfswissenschaftler** (Student Research Assistant) —
Medizinische Hochschule Hannover, Germany

• Developing an LLM- and Knowledge Graph-aware RAG based multi-agent system for scientific hypothesis generation from heterogeneous biomedical evidence; supervised by Dr. Zahra Ahmadi.

• Investigating evidence-based reasoning, calibrated confidence, and risk-controlled mechanisms for AI-assisted scientific reasoning under incomplete or heterogeneous evidence.

Mar '26 – May '26 **Student Hilfswissenschaftler** (Student Research Assistant) —
TU Braunschweig, Institute for Statistics and Foundations of Machine Learning, Germany

• Conducted research on calibration and uncertainty estimation in modern neural networks under the supervision of Prof. Dr. Nicole Mücke.

Publications

Accepted Papers

1. **S. Basak**, *Is grokking a loss of normal hyperbolicity of the interpolation manifold?*, 2026, Oral presentation at SKILL 2026 (Studierendenkonferenz Informatik), INFORMATIK 2026, Gesellschaft für Informatik e.V., Dresden, 23 September. To appear in GI-Edition Lecture Notes in Informatics (LNI). URL: https://arxiv.org/abs/2608.14803

2. R. Younis, **S. Basak**, L. Chavez, and Z. Ahmadi, *From literature to hypotheses: An AI co-scientist system for biomarker-guided drug combination hypothesis generation*, 2026, Accepted at IEEE ICDM 2026, Demo track. Equal contribution. To appear in ICDM 2026 Workshop Proceedings. URL: https://arxiv.org/abs/2603.00612

Research Projects

Covariate Shift Regularization using Kernel Methods *(Gradient Descent, Linear Regression)*

Investigated the how covariate shift affects model generalization and predictive reliability under distribution mismatch; evaluated importance-weighting and self-attention-based approaches for adapting models to changing input distributions. Student project supervised by Prof. Dr. Nicole Mücke, graded **1.0**. **GitHub**: `dist_shift_regularization` .

Uncertainty Quantification in Neural Networks *(Neural Networks, Gaussian Process, Distribution Shift)*

Investigated uncertainty estimation in neural networks through the connection between finite-width networks and infinite-width Gaussian-process limits; compared uncertainty estimates from Neural Network Gaussian Processes (NNGPs), deep ensembles, MC dropout, and RBF Gaussian processes under distribution shift; studied how different methods behave when test distribution differs from training distribution. **GitHub**: `infinite-width-uq` .

Binary AMR Prediction from *PBP4* using Transformers *(DNABERT, Nucleotide Transformer, PyTorch)*

Benchmarked encoder-only pretrained transformer models, including DNABERT and Nucleotide Transformer, for binary antimicrobial resistance prediction using *pbp4* genomic sequences; investigated representation learning and parameter-efficient adaptation of pretrained biological language models using PyTorch. Seminar *"Deep Learning in Molecular Biology"*, supervised by Prof. Dr. Alice C. McHardy. **GitHub**: `AMRxformer` .

Teaching Experience

Apr '25 – Sep '25 **Student Hilfswissenschaftler** (Teaching Assistant) — *TU Braunschweig, Institut für Analysis und Algebra, Germany*

• Led tutorials and graded assignments for *Mathematical Algorithms and Programming 2* (SS 2025).

Industry Experience

Aug '24 – Feb '25 **Intern, Data Analytics & AI** — *Volkswagen Financial Services AG, Braunschweig, Germany*

• Built an ML pipeline automating invoice analysis and approval in vehicle maintenance workflows; reducing manual review time; designed a multi-stage Docker setup improving container security and pipeline build time.

Jun '18 – Oct '23 **Strategy Consultant** — *Wipro Ltd., Bangalore, India*

• Led AI/ML innovation scouting and startup partnerships, from technical evaluation through joint proof-of-concept development and go-to-market planning.

Mar '15 – Jun '16 **Solution Integrator** — *Ericsson Global India Services Pvt. Ltd., Kolkata, India*

• Configured and integrated a billing system for a leading US-based mobile carrier, improving end-to-end system consistency; analyzed historical usage patterns to design billing plans aligned with client offerings and user behavior.

Technical Skills

Mathematical ML	Statistical learning • Optimization • Generalization • Uncertainty quantification • Calibration • Distribution shift • Deep learning
Program & Scientific Computing	Python • NumPy • Pandas • SQL • Julia • scikit-learn • PyTorch • TensorFlow
Deep Learning	Hugging Face • Transformers • LoRA/PEFT
LLM Systems	Large Language Models (LLM) • Retrieval-Augmented Generation (RAG) • Multi-Agent Systems • Knowledge Graphs
Tools	Git • Docker • Linux • LaTeX
Spoken	English (proficient) • Bengali (native) • Hindi (proficient) • German (beginner)

Awards & Recognition

2025/26 **Deutschlandstipendium** — Merit-based scholarship for high-achieving and committed students.

2019, 2023 **Industry recognition, Wipro Ltd.** — Excellence Award (Q_3–Q_4, FY19); Customer Centricity Award (Q_1, FY23).